

# Solution Guide: Detailed Answers and Explanations

## 1. Total Number of Apps

**Visual:** Card

**Answer:**

Displays the **total number of apps** available in the Google Play Store dataset.

**Explanation:**

This KPI helps understand the overall size of the dataset and serves as a baseline for all subsequent analyses.

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## 2. Unique Categories of Apps

**Visual:** Card

**Answer:**

Shows the **number of distinct app categories**.

**Explanation:**

It identifies the diversity of app offerings available on the Google Play Store.

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## 3. Distribution of Apps by Size

**Visual:** Clustered Column Chart

**Answer:**

Apps are grouped into **Size (MB) bins**, and the chart displays how many apps fall into each size range.

**Explanation:**

This helps identify whether most apps are lightweight or require significant storage space.

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## 4. Free vs Paid Apps

**Visual:** Pie Chart

**Answer:**

Compares the count of apps based on the **Type** field (Free or Paid).

**Explanation:**

Typically, free apps dominate the Play Store, while paid apps constitute only a small percentage.

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## 5. Distribution of Ratings

**Visual:** Clustered Column Chart

**Answer:**

Shows the number of apps across different rating values.

**Explanation:**

This reveals whether most apps are highly rated, moderately rated, or poorly rated.

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## 6. Apps with Highest Installs

**Visual:** Table

**Answer:**

Lists app names alongside their total installations.

**Explanation:**

This helps identify the most popular apps based on user adoption.

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## 7. Another Total Apps KPI

**Visual:** Card

**Answer:**

Displays a summary measure related to the total number of apps.

**Explanation:**

Often used to reinforce key business metrics.

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## **8. Average Number of Reviews**

**Visual:** Card

**Answer:**

Shows the average reviews received per app.

**Explanation:**

Reviews indicate user engagement and popularity.

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## **9. Average App Size**

**Visual:** Card

**Answer:**

Displays the mean app size.

**Explanation:**

Provides insight into typical storage requirements of Play Store applications.

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## **10. Apps Updated in 2018**

**Visual:** Card

**Answer:**

Counts apps updated during 2018.

**Explanation:**

Measures developer activity and maintenance frequency.

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## 11. Relationship Between Installs and Ratings

**Visual:** Scatter Plot

**Answer:**

Compares total installs against ratings.

**Explanation:**

Helps determine whether highly rated apps tend to have more downloads.

**Observation:**

A positive relationship may indicate that better-rated apps attract more users.

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## 12. Average Rating Analysis

**Visual:** 100% Stacked Bar Chart

**Answer:**

Displays average ratings.

**Explanation:**

Allows comparison of rating performance across categories or segments.

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## 13. Price vs Rating

**Visual:** Scatter Plot

**Answer:**

Analyzes how app prices relate to ratings.

**Explanation:**

Determines whether expensive apps receive better ratings.

**Observation:**

High price does not necessarily guarantee higher ratings.

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## 14. Category-wise App Count by Rating

**Visual:** Clustered Bar Chart

**Answer:**

Shows app counts within categories segmented by ratings.

**Explanation:**

Helps identify categories with the highest concentration of highly rated apps.

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## 15. Genres with More Than 1 Million Installs

**Visual:** 100% Stacked Column Chart

**Answer:**

Identifies genres having apps exceeding one million installs.

**Explanation:**

Highlights genres that attract large audiences.

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## 16. Average Days Between Updates

**Visual:** Clustered Column Chart

**Answer:**

Displays the average number of days between updates for each app.

**Explanation:**

Frequent updates often indicate active developer support and bug fixes.

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## 17. Category Distribution by Rating Bins

**Visual:** 100% Stacked Bar Chart

**Answer:**

Shows how ratings are distributed within app categories.

**Explanation:**

Useful for comparing customer satisfaction across categories.

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## 18. Rating and Reviews by App

**Visual:** Matrix (Pivot Table)

**Answer:**

Lists each app with:

- Total Ratings
- Total Reviews

**Explanation:**

Enables detailed app-level comparison.

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## 19. Content Rating Distribution

**Visual:** 100% Stacked Bar Chart

**Answer:**

Shows the distribution of apps by content rating categories such as:

- Everyone
- Teen
- Mature 17+

**Explanation:**

Helps understand the intended audience of Play Store apps.

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## 20. Category-wise Install Analysis

**Visual:** Ribbon Chart

**Answer:**

Ranks categories based on total installs.

**Explanation:**

Ribbon charts highlight changes in category rankings.

**Observation:**

Categories like Games and Communication often dominate installations.

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## 21. App Performance Matrix

**Visual:** Pivot Table

**Answer:**

Displays:

- Installs
- Ratings
- Reviews

for each app.

**Explanation:**

Provides a comprehensive view of app performance metrics.

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## 22. App Updates Over Time

**Visual:** Line Chart

**Answer:**

Tracks the number of apps updated by year.

**Explanation:**

Shows trends in developer activity.

**Observation:**

An increasing trend suggests growing platform engagement.

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## 23. Ratings Across Install Ranges

**Visual:** Line Chart

**Answer:**

Compares ratings across installation bins.

**Explanation:**

Helps determine whether heavily downloaded apps receive better ratings.

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## 24. Install Category vs Ratings

**Visual:** Dual-Axis Line Chart

**Answer:**

Compares:

- Number of apps
- Total ratings

across install categories.

**Explanation:**

Highlights segments with both high popularity and strong user satisfaction.

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## 25. Genre-wise Ratings

**Visual:** Clustered Bar Chart

**Answer:**

Displays total ratings by genre.

**Explanation:**

Identifies genres receiving the highest user appreciation.

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## Overall Insights

- The Google Play Store is dominated by **free applications**.
- A few categories account for a significant portion of **total installs**.
- **Higher installs generally correspond with stronger ratings**, although exceptions exist.
- Genres attracting millions of installs represent major market opportunities.
- Frequent updates are associated with active maintenance and potentially better user experiences.